Wi-Fi Channel Selection Based on Urban Interference Measurement

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ABSTRACT

Increasing availability and usability enhancement of Wi-Fi in public areas has become more active. However, due to the dense deployment of Wi-Fi access points (APs), there is a chaotic and disorderly environment in urban areas. In our previous work, we have designed a function that predicts the network performance at each Wi-Fi AP according to the measurement of IEEE802.11 MAC frames sensed in each Wi-Fi channel. However, it was not examined in such scenarios assuming urban environment. We should understand the situations of current Wi-Fi AP deployment and traffic conditions, and should confirm the effectiveness of channel migration in such realistic environment. In this study, we proposed urban Wi-Fi channel utilization model based on real urban Wi-Fi measurement. We show that our method can predict the best channels and APs can migrate to them in the urban scenario.

CCS Concepts

•**Networks** → **Network management;** *Network simulations; Network performance analysis;*

Keywords

2.4GHz Wi-Fi, Interference, Channel Selection, Machine Learning

1. INTRODUCTION

It has continued to increase in foreign tourists because Japan is the host country for the Tokyo Olympic and Paralympic in 2020. Since foreign tourists tend to use Wi-Fi to avoid the monetary cost of using LTE, the Ministry of Internal Affairs and Communications issued an action plan, called "SAQ2 JAPAN Project", which promotes Wi-Fi communication environment improvement and realizes the world's highest level ICT environment in June 2014. Softbank also provided nationwide 400,000 APs for foreigners. The movement toward increasing availability and usability enhancement of Wi-Fi in public areas have become more active. In addition, Wi-Fi has also been important as alternative infrastructure of mobile networks in the event of disaster and low-cost smart city foundation. In Barcelona, the information about the urban infrastructure such as street light management, human flow, and smart parking, and the environment like temperature, air quality, and noise levels is aggregated through a Wi-Fi-based platform. Also in the field of Intelligent Transportation System (ITS), applying Wi-Fi infrastructure to Roadside-to-Roadside (R2R) and Roadside-to-Vehicle (R2V) communication has been studied. Roadside units share the traffic information and feed this information back to vehicles through Wi-Fi communication in future systems. Therefore Wi-Fi becomes one of the most important infrastructures and are expected to be utilized in various environments.

On the other hand, for example, as shown in Figure 1, the number of Wi-Fi APs, STAs and devices is steadily increasing. Especially in the center of the population concentration city, there is a chaotic and disorderly environment due to the dense deployment of public/private Wi-Fi, proliferation of mobile routers and the increase of multi-band Wi-Fi chipsets and ITS vehicle-mounted devices. For such a dense Wi-Fi problem, the IEEE802.11ax task group reports that Wi-Fi throughput can be nearly doubled using Dynamic Sensitivity Control (DSC) and Transmit Power Control (TPC) [10]. Meanwhile, our goal is autonomous and efficient frequency reuse of each AP which has the existing architecture like IEEE802.11a/g/n. For this purpose, we have developed a channel migration technology that can control congestion among Wi-Fi channels based on the concept of interference environment sensing. In this approach, based on the highly-precise Wi-Fi simulation and machine learning, we have proposed a function that predicts the network performance and the interference level at each Wi-Fi AP from the observation of IEEE802.11 MAC frames in each Wi-Fi channel. It is possible to indicate a channel that is expected to provide the highest quality [4]. However, it was not examined in such scenarios assuming urban environment. We should understand the situations of current Wi-Fi AP deployment

MOBIQUITOUS '16, November 28-December 01, 2016, Hiroshima, Japan

ISBN 978-1-4503-4750-1/16/11...$15.00

DOI: http://dx.doi.org/10.1145/2994374.2994402

Figure 1: Wi-Fi APs around Osaka Station [12]

and traffic conditions, and should confirm the effectiveness of channel migration in such realistic environment.

In this paper, we propose urban Wi-Fi channel utilization model based on real urban Wi-Fi measurement. To build this urban model, we surveyed urban areas like in Osaka (Figure 1). We conducted Wi-Fi frame monitoring with AirPcap [2] at 10 locations including shopping malls, cafes, commercial buildings and stations on both weekdays and holidays. This monitoring was carried out in each channel for five minutes, and we got the number of APs and STAs from the source and destination MAC addresses of Beacon frames and Probe Response frames. Then we analyzed this actual measurement and added more traffic considering the future growth of Wi-Fi traffic. This future traffic prediction is determined based on the distribution of the number of APs in each channel. Finally, we demonstrate that the proposed function can be used in this urban model for channel selection at each AP. We show that this function can specify the best channels and APs can increase the throughput compared with other channels.

2. RELATED WORK

Due to autonomous behavior of APs in uncoordinated environment and difficulty of performance estimation, Wi-Fi channel selection is not an easy problem. However, a large amount of effort has been dedicated to tackling this problem in different ways. Channel hopping, which has been incorporated into Bluetooth or some other systems, is a potential solution. Nevertheless, it basically causes a certain overhead in hopping, and in particular, Wi-Fi system is not designed for frequent hopping among channels due to its association overhead between APs and STAs. Although some work like [8] considers dynamic channel sequences, it needs channel status estimation by monitoring. Different from such solutions, parameter control (such as adaptive carrier sense threshold control and transmission rate/power control) aims at avoiding congestion spatially. From the research results in [3], [7] addresses the fact that the transmission power of most Wi-Fi APs are configured to maximum in the factory settings, which often induces unnecessary interference, but self-control of transmission power by APs may cause unidirectional links. Therefore, cross-layer control is recommended where the carrier-sense threshold is coordinately controlled with transmission power (e.g. the transmission power of APs with heavy traffic load should be larger). We note that these approaches can co-work with any channel selection strategy as it may mitigate interference. Finally, traffic monitoring (RSS monitoring in particular) is often utilized for channel quality estimation. For example, [1] has pointed out SNR and RSS do not provide sufficient information to estimate L2 performance, but recent work [9] presents a novel method to accurately identify the presence of non-Wi-Fi machines using information obtained through off-the-shelf Wi-Fi cards. In addition, [6] presents an approach of estimating frame collision and loss rates in IEEE802.11MAC. It employs probabilistic models to infer backoff occurrence due to carrier-sense operations. Besides, [5] presents a distributed channel selection algorithm and an AP selection strategy for STAs. However, the goal of this approach is fairness among STAs while ours is selection of a channel with least interference effect.

3. URBAN WI-FI MODEL

3.1 Wi-Fi MAC Frame Capture in Osaka Downtown

For the reproduction of Wi-Fi overcrowded environment in simulation, we conducted frame monitoring. As mentioned above, we used a Windows-based USB wireless traffic analyzer, *AirPcap Nx* shown in Figure 2. *AirPcap Nx* can capture 802.11 MAC frames including frames that Frame Check Sequence (FCS) is not correct. We captured the frames that are sent and received at APs and STAs in each channel for five minutes. From this observation, we obtained some statical values such as the number of APs and STAs in each channel, traffic volume and RSSI value of all sensed frames. To obtain the number of APs, we counted the source MAC addresses of Beacon frames and the destination MAC addresses of Probe Response frames. In the same way, to obtain the number of STAs, we counted the destination MAC addresses of Probe Response frames. The volume of traffic is calculated by averaging the sum of the frame length transmitted and received in each channel, and RSSI is similarly calculated by averaging RSSI of all frames in each channel. So we did not use the private information like contents of data frames.

Figure 3 shows the ten locations where we performed Frame Monitoring in Umeda. As Umeda station is reported the busiest train station in Osaka, Umeda is a typical urban environment. These 10 locations include stations (green boxes in Figure 3), shopping malls (black), cafes (blue) and office buildings (orange) where mobile routers and non-Wi-Fi devices can be seen. We surveyed every location on both weekdays and holidays to find the long-term difference.

3.2 Modeling Urban Environment

In order to reflect the characteristics of Wi-Fi utilization, we have analyzed the obtained dataset. We focused on the distribution of APs over 13 channels at each loca-

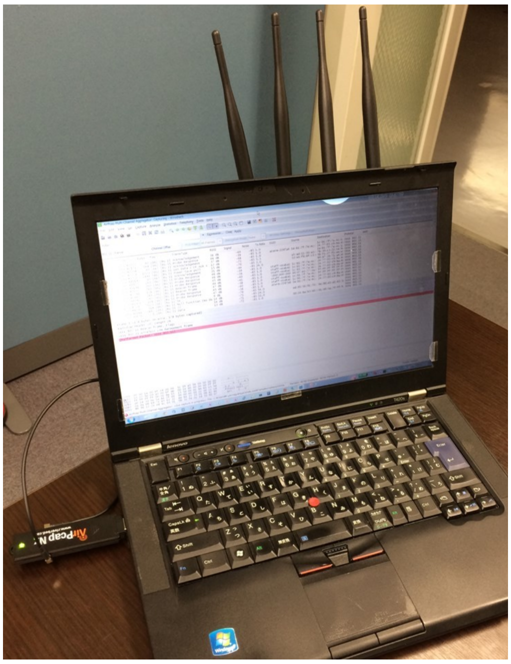


Figure 2: *AirPcap Nx*

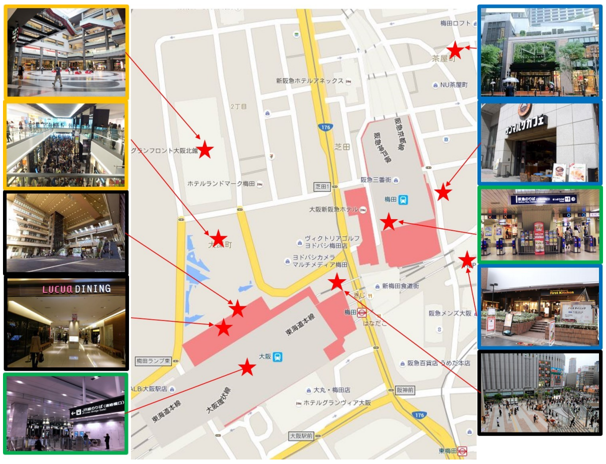


Figure 3: Frame Monitoring Locations

tion. It is easily imagined that most APs exist either of channels 1, 6 and 11. At first, to find various distribution patterns, we applied k-means clustering to this dataset. K-means is a typical and simple clustering method for unsupervised learning. We apply k-means algorithm to the dataset captured on holidays and find three distribution patterns, a) default, b) coordinated, c) uncoordinated in Figure 4. The default pattern can be seen at shopping malls and shows the ratio of channels 1 and 2 is extremely high. We can consider that this case is the result of easy installation by default settings. The coordinated pattern can be seen in office buildings and shopping malls and shows some peaks at channels 1, 6 and 11. We can consider that such buildings avoid interference with each other. The uncoordinated pattern can often be seen outdoors and at busy areas. This shows the number of APs in channels 1 and 11 is slightly higher than the others. This is the most chaotic situation.

Table 1: Correlation of AP distributions (Holidays vs Weekdays)

Location	Correlation
commercial building 1	0.82
commercial building 2	0.78
shopping mall 1	0.75
shopping mall 2	0.91
shopping mall 3	0.73
station 1	0.86
station 2	0.78
cafe 1	0.80
cafe 2	0.77
cafe 3	0.56

Table 2: Correlation of AP and STA distributions

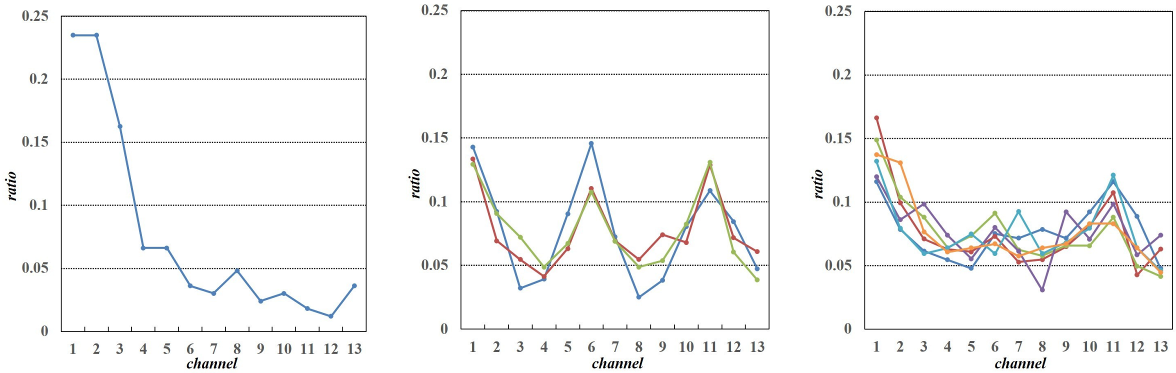
Location	Correlation
commercial building 1	0.95
commercial building 2	0.92
shopping mall 1	0.91
shopping mall 2	0.85
shopping mall 3	0.93
station 1	0.63
station 2	0.92
cafe 1	0.74
cafe 2	0.72
cafe 3	0.53

Secondly, we compared two AP distributions of holidays and weekdays. We assume that two distributions of holidays and weekdays at the same location have a strong correlation. To understand this correlation, we calculated correlation coefficients shown in Table 1. As shown in Table 1, except the cafe 3 case, coefficients are above 0.70. This means two distributions have a high correlation. From this result and for simplicity, we consider the distribution of STAs over channels is the same as APs.

Thirdly, we compared two distributions of APs and STAs in the same location. In the same way, we assume that these two distributions have a strong correlation because STAs need to associate with APs. To clarify this correlation, we calculated correlation coefficients shown in Table 2. As shown in Table 2, except the station 1 and cafe 3, coefficients are above 0.7. For simplicity, we also consider the distribution of STAs is the same as APs.

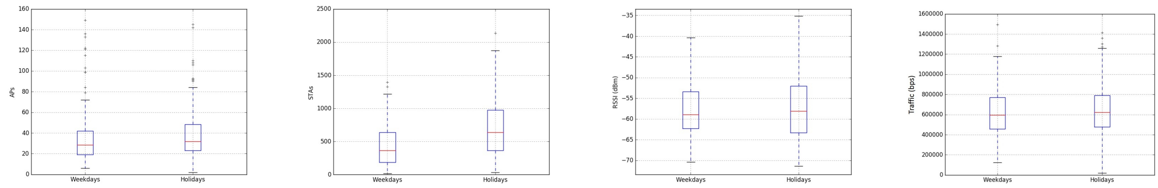
Finally, we compared the total sums of APs, STAs and the average of RSSI, traffic volume on holidays and weekdays. This result is shown Figure 5. Except for Figure 5 (b), we find that the mean values do not really change but the variances change slightly. As shown in Figure 5 (b), the mean of the number of STAs increases 60% on holidays. This is because more people use their laptops in public areas like cafes on holidays. We need to consider that the traffic gain on holidays this is part of our future work.

3.3 Urban Interference Problems



(a) default (b) coordinated (c) uncoordinated

Figure 4: Normalized AP Distribution in each channel and each location



(a) APs (b) STAs (c) RSSI (d) Traffic

Figure 5: Difference of these total volume between Weekdays and Holidays

By analyzing our urban model, we find typical urban interference problems. As seen in Figure 4(c), especially in urban outdoor environments, it is found that uncoordinated AP installation is underway. Therefore, we investigate problems under these situations. We prepare a 200m×200m field in the simulator, called *Scenargie*, and 100 AP-STA pairs are deployed randomly. Their traffic settings are set to the same volume and their channels are determined by the distribution (a), (b) and (c) in Figure 4. We show these three simulation results in Figure 6. The delay means the average value of all AP-STA pairs. From this result, (b) coordinated case is the best AP distribution to avoid the interference. This fact indicates importance to select interference-free channels as the scenario (b) models administrated environments.

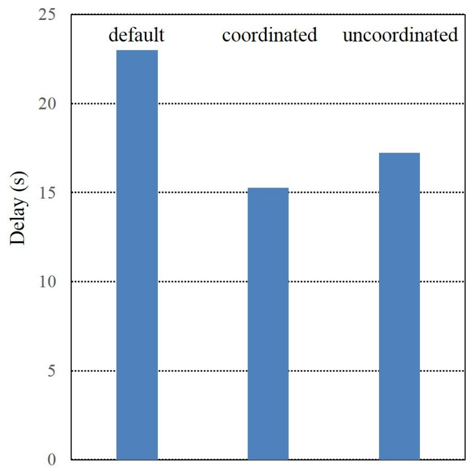


Figure 6: Delay Values in each situation((a) defalut, (b) coordinated, (c) uncoordinated)

4. APPLICATION OF URBAN SCENARIOS TO WI-FI CHANNEL SELECTION

4.1 Wi-Fi Channel Selection Strategy Overview

Let AP denote an IEEE802.11g AP of interest (called *target* AP) and ST denote each Wi-Fi STA which is associated with AP. Suppose AP (and its STAs) currently uses channel c_{cur} in the Wi-Fi channel set (denoted as C) and monitors the traffic of other AP(s) (interference AP(s)) that use the same or other channels. In particular, AP calculates the following two values for each channel k; (i) temporal channel utilization ratio (or simply channel utilization ratio) denoted as $t(k)$, and (ii) their received signal strength denoted as $s(k)$, respectively. According to [4], we will use normalized ones as follows. $t(k)$ is defined as follows.

$$t(k) = \frac{ave\_bitrate(k)}{data\_rate} + num\_frames(k) \cdot T_{preamble} \quad (1)$$

The average bitrate can basically be obtained by counting the bytes of all the MAC frames on channel k, and the data rate of IEEE802.11g is 6, 9, 12, 18, 24, 36, 48 or 54Mbps (in case of OFDM PHY). In addition, $t(k)$ is

Table 3: Parameter Settings for Dataset Generating

Parameter	Values
Area Size	150m×150m
Interference AP - Target AP Distance	[20m, 400m] step=20m
L7 traffic (Interference AP)	IPERF [1Mbps, 9Mbps] step=0.5Mbps
L7 traffic (Target AP)	IPERF [1Mbps. 9Mbps] step=1Mbps
IEEE802.11g Data Rate	9Mbps (BPSK 3/4)
Channels	$c_{\text{new}} = 6$, $c_{\text{inf}} \in \{6, 7, 8, 9\}$

corrected based on the total number of received frames and the duration of the preamble. Due to CSMA/CA features and inter-frame spacing, $t(k)$ cannot be 1.0, but a larger value means higher utilization. $s(k)$ is defined as follows.

$$s(k) = \frac{ave\_rss(k) - \theta_{\min}}{\theta_{\max} - \theta_{\min}} \quad (2)$$

where $\theta_{\min}$ and $\theta_{\max}$ represent the minimum RSS threshold of data frame reception (-90dBm in IEEE802.11g) and expected maximum RSS (usually -40dBm or around), respectively.

Then our channel selection strategy is designed as follows. Given c_{new} to which AP likes to switch from c_{cur} and given observations $t(c_{\text{inf}})$, $s(c_{\text{inf}})$ and $t(c_{\text{cur}})$, we provide two functions $f_{\text{D}}(c_{\text{new}}, t(c_{\text{inf}}), s(c_{\text{inf}}), t(c_{\text{cur}}))$ and $f_{\text{T}}(c_{\text{new}}, t(c_{\text{inf}}), s(c_{\text{inf}}), t(c_{\text{cur}}))$, which return the expected L2 delay and (normalized) L2 throughput (*i.e.* channel utilization ratio), respectively. Having these two estimators, AP can predict the performance when it moves from c_{cur} to c_{new}, just by observing IEEE802.11 MAC frames and their RSS in channel c_{inf} at AP.

4.2 Dataset and Model for Estimator

In order to design accurate f_{D} and f_{T}, our basic policy is to use a large dataset, each of which shows relation between the observed $t(c_{\text{inf}})$, $s(c_{\text{inf}})$ and $t(c_{\text{cur}})$ and the corresponding delay/throughput to reveal the performance-observation relations and trends.

Such a dataset is generally hard to obtain in the real world because the number of scenarios is too many (in this paper we have prepared 12,960 as a total) and for each scenario, we have to set the real equipment. Instead, we rely on the highly-accurate commercial simulator (*Scenargie* 1.8 [11]) as mentioned earlier. Since it has an accurate OFDM subchannel spectrum spread model and complete and reliable implementation of the IEEE802.11 family, the simulation results are sufficiently dependable. The simulation settings are summarized in Table 3.

From our preliminary experiment, we can decide that the saturation points are around delay=0.1s because we observed delay values rose sharply after this point. As the delay and throughput are independent of interference traffic before reaching the saturation points and on the contrary, it is largely affected after the saturation point, it seems essential to use different functions before and after the saturation points, rather than using a single function. We have also examined several regression functions to represent the saturated-part using best-fit functions. Consequently, we have determined to use the following function with unknown parameters ($c_{i,j}$) as well as an unknown predicate $sat(t(c_{\text{inf}}), s(c_{\text{inf}}), t(c_{\text{cur}}))$.

$$f_{\text{D}}(c_{\text{new}}, t(c_{\text{inf}}), s(c_{\text{inf}}), t(c_{\text{cur}})) = \begin{cases} t(c_{\text{cur}}) & (\textbf{if } |c_{\text{new}} - c_{\text{inf}}| > 3 \\ & \quad \textbf{or } \neg sat(t(c_{\text{inf}}), s(c_{\text{inf}}), t(c_{\text{cur}}))) \\ u_0 + u_1 \log(t(c_{\text{inf}}) + t(c_{\text{cur}})) + u_2 \cdot t(c_{\text{inf}}) + u_3 \cdot s(c_{\text{inf}}) + u_4 \cdot t(c_{\text{cur}}) & (\textbf{elsif } c_{\text{new}} == c_{\text{inf}}) \\ v_0 + v_1 \cdot t(c_{\text{inf}}) + v_2 \cdot s(c_{\text{inf}}) + v_3 \cdot t(c_{\text{new}}) + v_4 \cdot t(c_{\text{inf}}) \cdot s(c_{\text{inf}}) + v_5 \cdot s(c_{\text{inf}}) \cdot t(c_{\text{new}}) + v_6 \cdot t(c_{\text{inf}}) \cdot t(c_{\text{new}}) + v_7 \cdot t(c_{\text{inf}}) \cdot s(c_{\text{inf}}) \cdot t(c_{\text{new}}) & (\textbf{otherwise}) \end{cases} \quad (3)$$

The predicate *sat* is a binary classifier to determine whether the traffic in channel c_{cur} causes saturation in the new channel c_{new} or not. The first term models the case that the performance is not affected by traffic in channel c_{inf} due to channel separation or enough space to accommodate $t(c_{\text{cur}})$.

If it is affected (*i.e.* if it is expected to cause saturation of capacity) and if interference APs reside in new channel c_{new}, then we include a logarithmically curved function. As all the frames are on the same channel, we introduce this function to represent performance degradation of CSMA/CA-based systems as a increase of the sum of interference and own traffic.

Finally, if it is not any of the above, then we employ another general function to find the interference effect between neighboring channels.

4.3 Determining Classifier and Model Parameters

Firstly, in order to obtain *sat* predicate, we have applied Support Vector Machine (SVM) based learning. We have labelled '*saturated*' or '*unsaturated*' to each data in the dataset. This decision is simply done by the delay values where 100ms delay is considered the saturation point. Then using the labelled delay/throughput with a vector ($t(c_{\text{inf}})$, $s(c_{\text{inf}})$, $t(c_{\text{cur}})$) as a training dataset, we finally obtained an SVM classifier, which is directly used as *sat* function. Secondly, we have applied multiple regression analysis to determine all the unknown parameters (u_i and v_j, $0 \leq i \leq 4$ and $0 \leq j \leq 7$) to model the performance after the saturation points.

4.4 Estimator Validation

We show the capabilities of our functions f_{D} and f_{T} to estimate the delay and throughput. We have prepared additional 2,592 scenarios randomly, following the configurations are shown in Table 3. However, we have carefully generated this dataset for testing so that training and test datasets are mutually disjoint.

Firstly, we show the classification results in Table 4. We show four subtables (confusion matrices) categorized by channel distance $|c_{\text{new}} - c_{\text{inf}}|$=0, 1, 2 or 3. We note

Table 4: ***sat*** **classifier output with different channel distance $|c_{\text{new}} - c_{\text{inf}}|$**

(a) 0

	T	F
T	371	0
F	5	272

(b) 1

	T	F
T	346	0
F	15	287

(c) 2

	T	F
T	314	1
F	12	321

(d) 3

	T	F
T	265	0
F	10	373

Table 5: Coefficients of Determination of f_{D} and f_{T} with different channel distance $|c_{\text{new}} - c_{\text{inf}}|$

Channel Distance	Co. of Det. (R^2)	
	f_{D}	f_{T}
0	0.8215	0.9815
1	0.7906	0.7520
2	0.8463	0.8161
3	0.8306	0.8796

that the rows show the groundtruth and the columns show *sat* outputs. From this result, the classification error is rather false positive, but even in the worst case, the error was 2.3%. The average error was 1.7%, both of which are negligible values.

Secondly, we have examined the accuracy of regression functions. As we have used regression analysis, we may directly refer to the coefficient of determination (R^2) to validate the fitting to the values. We show the results in Table 5. We note that R^2 values closer to 1.0 are better. As seen in the table, in all the cases, the model well captures the delay and throughput behavior as they are almost close to 0.8 or larger. In particular, the throughput with $|c_{\text{new}} - c_{\text{inf}}| = 0$ is the best case where 0.98 is achieved.

Finally, we have shown the accuracy of the delay and throughput estimation results. We have also summarized the average mean square errors in Table 6. We can see that distance 0 is the most accurate case. For other cases, the error of delay estimation is about 1 second and the error of throughput estimation is 10% or less. These values are quite reasonable considering the fact that we only use MAC frame passive observation and this is a lightweight estimation function that can easily be implemented on any APs as a value-added function. For visualization purpose, we have also shown the graphs in Figure 7 and 8 that show both groundtruth and the corresponding estimation results where the plots are sorted by the groundtruth values.

4.5 Effect from Multiple Channels

In order to consider the effect from multiple channels, we have prepared multi-channel scenario shown in Figure 9. There are two interference AP-STA pairs and they use different channels. One of interference pairs uses the same channel 0 with *AP*, and the other uses the channel from 1 to 3. The traffic configurations of *AP* and two interference pairs are IPERF 9Mbps, 3Mbps, and 3Mbps respectively. The results are shown in Figure 10. We decide to apply weighted sum for our function to merge the impact from multiple channels. We empirically determine the weighted coefficient is 1, 1/2, 1/4, 1/8 when the channel distance is 0, 1, 2, 3, respectively. In this case, the function outputs are shown in Table 7. The correlation coefficient is above 0.97, which means this function can capture the tendency of multi-channel effect.

Table 6: Average Mean Square Errors of f_{D} and f_{T} with different channel distance $|c_{\text{new}} - c_{\text{inf}}|$

Channel Distance	Ave. Mean Square Errors	
	f_{D} (sec.)	f_{T} (ratio)
0	0.1759	0.0178
1	1.1545	0.1055
2	0.9894	0.0897
3	1.1358	0.0879

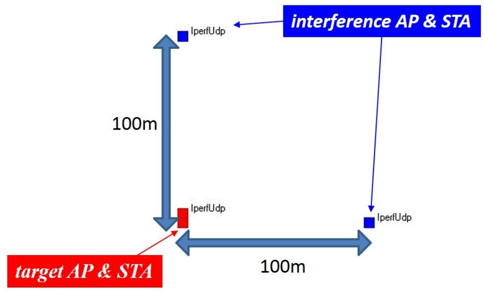
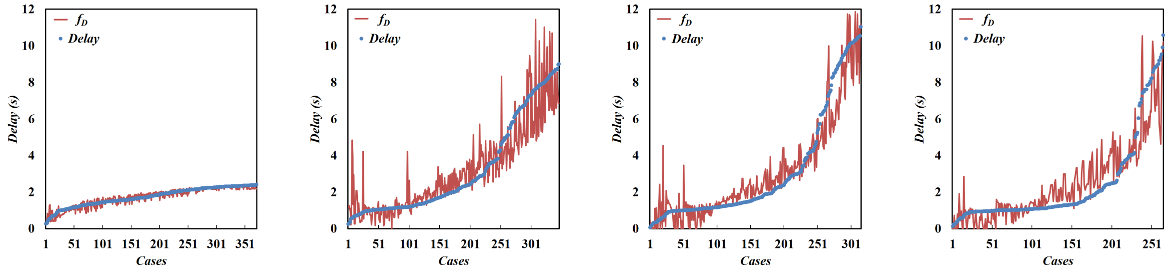


Figure 9: Multi-Channel Scenario

5. EVALUATION OF ESTIMATOR

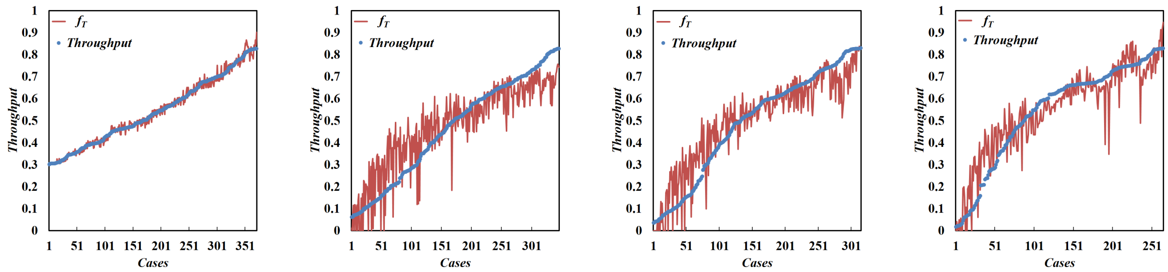
In order to evaluate our proposed estimator in realistic scenario, we have prepared the evaluation environment like urban situation, shown in Figure 11. We have prepared a representative interference AP-STA pair in each channel, and these 13 pairs are deployed in a position away about 100m from the target AP-STA pair. Their traffic configurations are adjusted to IPERF 5Mbps (channel 1, 6, 11), 1Mbps (channel 3, 4, 13) and 3Mbps (others). The target AP-STA pair communicates from channel 1 to 13 with IPERF 4.5Mbps. We compare the estimator output and groundtruth and confirm that the estimator can capture the best channel.

Figure 12 shows the delay value (groundtruth) of target pair in each channel. From this figure, it is found that the target pair should select channel 13. In addition, Figure 13 shows the network quality of all pairs is the best when the target pair select channel 13. That is, we want to confirm that our proposed estimator can choose channel 13. The outputs of our proposed estimator are shown in Table 8. This shows our estimator can imply the best channel is 13 because the output is the lowest value. We confirmed the correlation coefficient is above 0.85 and out estimator can capture the tendency of groundtruth.



(a) 0 (b) 1 (c) 2 (d) 3

Figure 7: f_D (delay) output and groundtruth with different channel distance $|c_{new} - c_{inf}|$



(a) 0 (b) 1 (c) 2 (d) 3

Figure 8: f_T (throughput) output and groundtruth with different channel distance $|c_{new} - c_{inf}|$

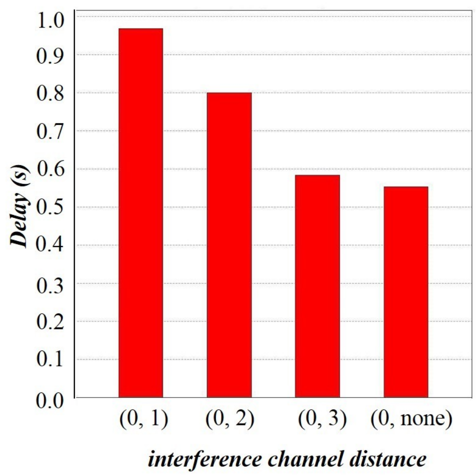


Figure 10: Delay Values in Multi-Channel Scenarios

Table 7: Function output, Groundtruth and Correlation in Multi-Channel Scenarios

Channel Distance	Function output	Groundtruth
(0,1)	1.417	0.968
(0,2)	1.125	0.800
(0,3)	0.879	0.584
(0,none)	0.671	0.554
Correlation		0.979

6. CONCLUSION

In this paper, we have built urban Wi-Fi utilization models from field Wi-Fi frame monitoring result carried out in Osaka downtown. To build more realistic model, we focused on the distribution of the number of APs over channels at each location. Based on this model, we conducted simulation experiments to understand a typical problem in urban areas.

We found that our channel selection function can predict the best channel and APs can migrate to the best channel. To evaluate our function, we conducted the simulation-based evaluation. We confirmed that the correlation coefficient is above 0.85, which means our function can capture the tendency of the real channel quality.

Acknowledgement

This research and development work was supported by the Strategic Information and Communications R&D Promotion Programme (SCOPE) No. 165007009 of the Ministry of Internal Affairs and Communications in Japan.

7. REFERENCES

[1] D. Aguayo, J. Bicket, S. Biswas, G. Judd, and R. Morris. Link-level measurements from an 802.11b mesh network. In *Proc. of ACM 2004 Conf. on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM 2004)*, pages 121–132, 2004.

[2] AirPcap. Riverbed. http://www.riverbed.com/.

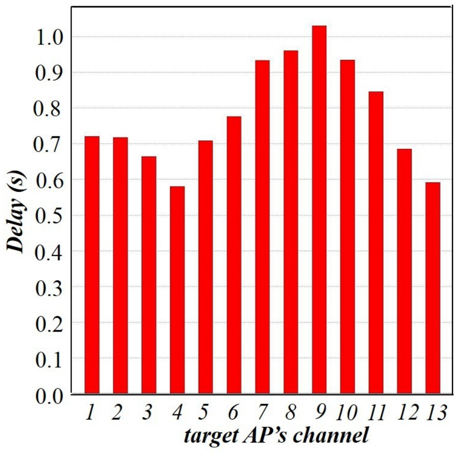


Figure 13: Evaluation Result of all APs

Table 8: Function output, groundtruth and Correlation in Evaluation Scenario

Channel	Function output	Groundtruth
1	3.684	2.133
2	3.632	2.224
3	3.352	1.788
4	3.606	1.634
5	4.569	2.009
6	5.869	3.125
7	6.245	4.993
8	6.043	5.484
9	6.043	6.082
10	6.246	3.889
11	5.868	3.743
12	4.315	2.063
13	2.592	1.635
Correlation		0.853

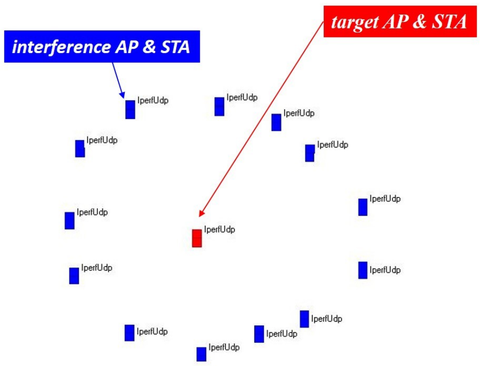


Figure 11: Evaluation Scenario

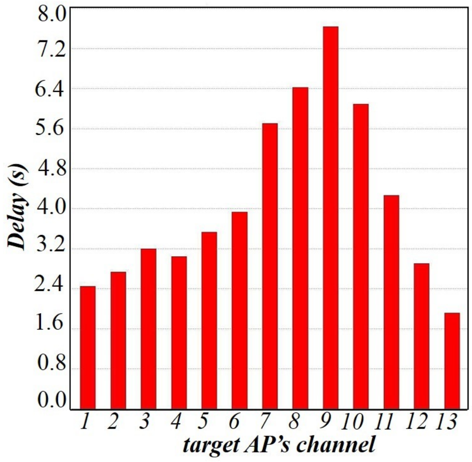


Figure 12: Evaluation Result of target AP

[3] A. Akella, G. Judd, S. Seshan, and P. Steenkiste. Self-management in chaotic wireless deployments. In *Proc. of ACM 11th Annual Int. Conf. on Mobile Computing and Networking (MobiCom 2005)*, pages 185–199, 2005.

[4] S. Kajita, H. Yamaguchi, T. Higashino, H. Urayama, M. Yamada, and M. Takai. Throughput and delay estimator for 2.4ghz wifi aps: A machine learning-based approach. In *Proc. 8th IFIP Wireless and Mobile Networking Conference (WMNC 2015)*, pages 223–226, 2015.

[5] B. Kauffmann, F. Baccelli, A. Chaintreau, V. Mhatre, K. Papagiannaki, and C. Diot. Measurement-based self organization of interfering 802.11 wireless access networks. In *Proc. of 26th IEEE Int. Conf. on Computer Communications (INFOCOM 2007)*, pages 1451–1459, May 2007.

[6] D. Malone, P. Clifford, and D. Leith. Mac layer channel quality measurement in 802.11. *Communications Letters, IEEE*, 11(2):143–145, Feb 2007.

[7] V. Mhatre, K. Papagiannaki, and F. Baccelli. Interference mitigation through power control in high density 802.11 WLANs. In *Proc. of 26th IEEE Int. Conf. on Computer Communications (INFOCOM 2007)*, pages 535–543, May 2007.

[8] A. Mishra, V. Shrivastava, D. Agrawal, S. Banerjee, and S. Ganguly. Distributed channel management in uncoordinated wireless environments. In *Proc. of ACM 12th Annual Int. Conf. on Mobile Computing and Networking (Mobicom 2006)*, pages 170–181, 2006.

[9] S. Rayanchu, A. Patro, and S. Banerjee. Airshark: detecting non-WiFi RF devices using commodity WiFi hardware. In *Proc. of 2011 Internet Measurement Conference (IMC 2011)*, pages 137–154, 2011.

[10] K. Shin, I. Park, J. Hong, D. Har, and D.-H. Cho. Per-node throughput enhancement in wi-fi densenets. *IEEE Communications Magazine*, 53(1):118–125, 2015.

[11] Space-Time Engineering, LLC. Scenargie. http://www.spacetime-eng.com/.

[12] WiGLE.NET. WiGLE. https://wigle.net/.